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Lower-Division Requirements

Foundations of Data Science:

Course	Description	Units	Prerequisites
Data 8 (COMPSCI/STAT/INFO C8): Foundations of Data Science	The UC Berkeley Foundations of Data Science course combines three perspectives: inferential thinking, computational thinking, and real-world relevance. Given data arising from some real-world phenomenon, how does one analyze that data so as to understand that phenomenon? The course teaches critical concepts and skills in computer programming and statistical inference, in conjunction with hands-on analysis of real-world datasets, including economic data, document collections,	4	None

	geographical data, and social networks. It delves into social issues surrounding data analysis such as privacy and design.		
STAT 135* + [CS 61A or ENGIN 7] *considered via petition if completed prior to Spring 18	**STAT 135: Concepts of Statistics** - A comprehensive survey course in statistical theory and methodology. Topics include descriptive statistics, maximum likelihood estimation, non-parametric methods, introduction to optimality, goodness-of-fit tests, analysis of variance, bootstrap and computer-intensive methods and least squares estimation. The laboratory includes computer-based data-analytic applications to science and engineering. **CS 61A: The Structure and Interpretation of Computer Programs** - An introduction to programming and computer science focused on abstraction techniques as means to manage program complexity. Techniques include procedural abstraction; control abstraction using recursion, higher-order functions, generators, and streams; data abstraction using interfaces, objects, classes, and generic operators; and language abstraction using interpreters and macros. The course exposes students to programming paradigms, including functional, object-oriented, and declarative approaches. It includes an introduction to asymptotic analysis of algorithms. There are several significant programming projects. **ENGIN 7: Introduction to Computer Programming for Scientists and Engineers** - Elements of procedural and object-oriented programming. Induction, iteration, and recursion. Real functions and floating-point computations for engineering analysis. Introduction to data structures. Representative examples are drawn from mathematics, science, and engineering. The course uses the MATLAB programming language. Sponsoring departments: Civil and Environmental Engineering and Mechanical Engineering.	4 4 4	**STAT 135:** STAT 134 or STAT 140; and MATH 54, EL ENG 16A, STAT 89A, MATH 110 or equivalent linear algebra. Strongly recommended corerequisite: STAT 133 **CS 61A:** MATH 1A (may be taken concurrently); programming experience equivalent to that gained in 3 or the Advanced Placement Computer Science A course. **ENGIN 7:** MATH 1B (may be taken concurrently)

Calculus I:

Course	Description	Units	Prerequisites
MATH 1A: Calculus	This sequence is intended for majors in engineering and the physical sciences. An	4	3.5 years of high school math, including

	introduction to differential and integral calculus of functions of one variable, with applications and an introduction to transcendental functions		trigonometry and analytic geometry, plus a satisfactory grade in one of the following: CEEB MAT test, an AP test, the UC/CSU math diagnostic test, or 32. Contact the Mathematics Department for details. Students with AP credit should consider choosing a course more advanced than 1A
Math 10A	The sequence Math 10A,B is intended for majors in the life sciences. Introduction to differential and integral calculus of functions of one variable. Representation of data, elementary probability theory, statistical models and testing. (F)	4, Students will receive 2 units for 10A after taking 1A	Three and one-half years of high school math, including trigonometry and analytic geometry. Consult the Mathematics Department for details. Students who have not had calculus in high school are strongly advised to take the Student Learning Center's Math 98 adjunct course for Math 10A; contact the SLC for more information.
Math 16A	Calculus of one variable; derivatives, definite integrals and applications, maxima and minima, and applications of the exponential and logarithmic functions. The sequence Math 16A, 16B is intended for students outside the physical sciences and engineering whose major requirements are satisfied by one or both courses, and who do not plan to take more advanced mathematics courses.	4, Students will receive no credit for 16A after taking 1A.	Three years of high school math, including trigonometry, plus a satisfactory grade in one of the following: CEEB MAT test, an AP test, the UC/CSU math diagnostic exam, or 32. Consult the mathematics department for details.

Calculus II:

Course	Description	Units	Prerequisites
MATH 1B: Calculus	Continuation of Math 1A. Techniques of integration; applications of integration.	4	MATH 1A or equivalent

	Infinite sequences and series. First-order ordinary differential equations. Second-order linear ordinary differential equations		
Math 10B	The sequence Math 10A,B is intended for majors in the life sciences. Elementary combinatorics and discrete probability theory. Introduction to graphs, matrix algebra, linear equations, difference equations, and differential equations.	4, Students will receive 2 units for 10B after taking 55	Math 10A or Math 1A
Math 16B	Continuation of 16A. Further techniques and applications of integration, Trigonometric functions, Differential equations, Probability and calculus, Sequences and series, Multivariable calculus.	4, Students will receive no credit for 16B after 1B.	16A

Linear Algebra:

Course	Description	Units	Prerequisites
MATH 54: Linear Algebra and Differential Equations	Basic linear algebra; matrix arithmetic and determinants. Vector spaces; inner product spaces. Eigenvalues and eigenvectors; linear transformations, symmetric matrices. Linear ordinary differential equations (ODE); systems of linear ODE. Fourier series.	4	MATH 1A-1B, MATH 10A-10B or equivalent
STAT 89: Linear Algebra for Data Science	An introduction to linear algebra for data science. The course will cover introductory topics in linear algebra, starting with the basics; discrete probability and how probability can be used to understand high-dimensional vector spaces; matrices and	4	Foundations of Data Science (COMPSCI C8 / INFO C8 / STAT C8) and one year of calculus

	graphs as popular mathematical structures with which to model data (e.g., as models for term-document corpora, high-dimensional regression problems, ranking/classification of web data, adjacency properties of social network data, etc.); and geometric approaches to eigendecompositions, least-squares, principal components analysis, etc.		
EL ENG 16A and EL ENG 16B: Designing Information Devices and Systems I & II	**EL ENG 16A: Designing Information Devices and Systems I** - This course and its follow-on course EE16B focus on the fundamentals of designing modern information devices and systems that interface with the real world. Together, this course sequence provides a comprehensive foundation for core EECS topics in signal processing, learning, control, and circuit design while introducing key linear-algebraic concepts motivated by application contexts. Modeling is emphasized in a way that deepens mathematical maturity, and in both labs and homework, students will engage computationally, physically, and visually with the concepts being introduced in addition to traditional paper/pencil exercises. The courses are aimed at entering students as well as non-majors seeking a broad foundation for the field. **EL ENG 16B: Designing Information Devices and Systems II** - This course is a follow-on to Electrical Engineering 16A, and focuses on the fundamentals of designing and building modern information devices and systems that interface with the real world. The course sequence provides a comprehensive introduction to core EECS topics in circuit design, signals, and systems in an application-driven context. The courses are delivered assuming mathematical maturity and aptitude at roughly the level of having completed Math 1A-1B, and are aimed at entering students as well as non-majors seeking a broad introduction to the field.	4 4	**EL ENG 16A:** MATH 1A, MATH 1B or equivalent (may be taken concurrently), CS 61A or equivalent (encouraged to be taken concurrently) **EL ENG 16B:** EL ENG 16A

Program Structures:

Course	Description	Units	Prerequisites
CS 61A: The Structure and Interpretation of Computer Programs	An introduction to programming and computer science focused on abstraction techniques as means to manage program complexity. Techniques include procedural abstraction; control abstraction using recursion, higher-order functions, generators, and streams; data abstraction using interfaces, objects, classes, and generic operators; and language abstraction using interpreters and macros. The course exposes students to programming paradigms, including functional, object-oriented,	4	MATH 1A (may be taken concurrently); programming experience equivalent to that gained in 3 or the Advanced Placement Computer Science A course.

	and declarative approaches. It includes an introduction to asymptotic analysis of algorithms. There are several significant programming projects.		
CS 88: Computational Structures in Data Science	Development of Computer Science topics appearing in Foundations of Data Science (C8); expands computational concepts and techniques of abstraction. Understanding the structures that underlie the programs, algorithms, and languages used in data science and elsewhere. Mastery of a particular programming language while studying general techniques for managing program complexity, e.g., functional, object-oriented, and declarative programming. Provides practical experience with composing larger systems through several significant programming projects.	2	Math 1A. Also, this course is a Data Science connector course and may only be taken concurrently with or after COMPSCI C8/INFO C8/STAT C8. Students may take more than one Data Science connector (88) course if they wish, concurrent with or after having taken the C8 course.
ENGIN 7: Introduction to Computer Programming for Scientists and Engineers	Elements of procedural and object-oriented programming. Induction, iteration, and recursion. Real functions and floating-point computations for engineering analysis. Introduction to data structures. Representative examples are drawn from mathematics, science, and engineering. The course uses the MATLAB programming language. Sponsoring departments: Civil and Environmental Engineering and Mechanical Engineering.	4	MATH 1B (may be taken concurrently)

Data Structures:

Course	Description	Units	Prerequisites
CS 61B: Data Structures	Fundamental dynamic data structures, including linear lists, queues, trees, and other linked structures; arrays strings, and hash tables. Storage management. Elementary principles of software engineering. Abstract data types. Algorithms for sorting and searching. Introduction to the Java programming language.	4	CS 61A or CS 88 or ENGIN 7

<u>Upper-Division Requirements</u>

Data 100: Principles and Techniques of Data Science:

Course	Description	Units	Prerequisites
DATA 100: Principles and Techniques of Data Science	Students in Data Science 100 will explore the data science lifecycle, including question formulation, data collection and cleaning, exploratory data analysis and visualization, statistical inference and prediction, and decision-making. The class focuses on quantitative critical thinking and key principles and techniques need to carry out this cycle. These include languages for transforming, querying and analyzing data; algorithms for machine learning methods including regression, classification and clustering; principles behind creating informative data visualizations; statistical concepts of measurement error and prediction; and techniques for scalable data processing.	4	DATA 8, CS 61A or CS 88, MATH 54
STAT 133 + STAT 135 + CS 61A	**STAT 133: Concepts in Computing with Data** - An introduction to computationally intensive applied statistics. Topics will include organization and use of databases, visualization and graphics, statistical learning and data mining, model validation procedures, and the presentation of results.	3	None
	STAT 135: Concepts of Statistics - A comprehensive survey course in statistical theory and methodology. Topics include descriptive statistics, maximum likelihood estimation, non-parametric methods, introduction to optimality, goodness-of-fit tests, analysis of variance, bootstrap and computer-intensive methods and least squares estimation. The laboratory includes computer-based data-analytic applications to science and engineering.	4	**STAT 135:** STAT 134 or STAT 140; and MATH 54, EL ENG 16A, STAT 89A, MATH 110 or equivalent linear algebra. Strongly recommended corerequisite: STAT 133
	CS 61A: The Structure and Interpretation of Computer Programs - An introduction to programming and computer science focused on abstraction techniques as means to manage program complexity. Techniques include procedural abstraction; control abstraction using recursion, higher-order functions, generators, and streams; data abstraction using interfaces, objects, classes, and generic operators; and language abstraction using interpreters and macros. The course exposes students to programming paradigms, including functional, object-oriented, and declarative approaches. It includes an introduction to asymptotic analysis of algorithms. There are several significant programming projects.	4	**CS 61A:** MATH 1A (may be taken concurrently); programming experience equivalent to that gained in 3 or the Advanced Placement Computer Science A course.

Computational & Inferential Depth:

Course	Description	Units	Prerequisites
CS 161: Computer Security	Introduction to computer security. Cryptography, including encryption, authentication, hash functions, cryptographic protocols, and applications. Operating system security, access control. Network security, firewalls, viruses, and worms. Software security, defensive programming, and language-based security. Case studies from real-world systems.	4	CS 61C (Machine Structures), plus either CS 70 (Discrete Mathematics) or MATH 55
CS 162: Operating Systems and Systems Programming	Basic concepts of operating systems and system programming. Utility programs, subsystems, multiple-program systems. Processes, interprocess communication, and synchronization. Memory allocation, segmentation, paging. Loading and linking, libraries. Resource allocation, scheduling, performance evaluation. File systems, storage devices, I/O systems. Protection, security, and privacy.	4	CS 61B, CS 61C, CS 70
CS 164: Programming Languages and Compilers	Survey of programming languages. The design of modern programming languages. Principles and techniques of scanning, parsing, semantic analysis, and code generation. Implementation of compilers, interpreters, and assemblers. Overview of run-time organization and error handling.	4	CS 61B, CS 61C
CS 168: Introduction to the Internet: Architecture and Protocols	This course is an introduction to the Internet architecture. We will focus on the concepts and fundamental design principles that have contributed to the Internet's scalability and robustness and survey the various protocols and algorithms used within this architecture. Topics include layering, addressing, intradomain routing, interdomain routing, reliable delivery, congestion control, and the core protocols (e.g., TCP, UDP, IP, DNS, and HTTP) and network technologies (e.g., Ethernet, wireless).	4	CS 61B, CS 162
CS 169: Software Engineering	Ideas and techniques for designing, developing, and modifying large software systems. Function-oriented and object-oriented modular design techniques, designing for re-use and maintainability. Specification and documentation. Verification and validation. Cost and quality metrics and estimation. Project team organization and management. Students will work in teams on a substantial programming project.	4	CS 61B, CS 61C, CS 70 or MATH 113
CS 170: Efficient Algorithms and	Concept and basic techniques in the design and analysis of algorithms; models of computation; lower bounds; algorithms for optimum search trees, balanced trees and UNION-FIND algorithms; numerical and algebraic algorithms; combinatorial	4	CS 61B, CS 70

Intractable Problems	algorithms. Turing machines, how to count steps, deterministic and nondeterministic Turing machines, NP-completeness. Unsolvable and intractable problems.		
CS 186: Introduction to Database Systems	Access methods and file systems to facilitate data access. Hierarchical, network, relational, and object-oriented data models. Query languages for models. Embedding query languages in programming languages. Database services including protection, integrity control, and alternative views of data. High-level interfaces including application generators, browsers, and report writers. Introduction to transaction processing. Database system implementation to be done as term project.	4	CS 61B, CS 61C
CS 188: Introduction to Artificial Intelligence	Ideas and techniques underlying the design of intelligent computer systems. Topics include search, game playing, knowledge representation, inference, planning, reasoning under uncertainty, machine learning, robotics, perception, and language understanding.	4	CS 61A, CS 61B, CS 70
ECON 140*: Economic Statistics and Econometrics *mutually exclusive with ECON 141 and STAT 135	Introduction to problems of observation, estimation, and hypothesis testing in economics. This course covers the linear regression model and its application to empirical problems in economics.	4	ECON 100A-100B or ECON 101A-101B or equivalent and Statistics 20, 21, 25, or 131A or equivalent; ECON 140, 141, and STAT 135 are mutually exclusive. Students may only take one of these three courses, and must select a second course from elsewhere on this list
ECON 141*: Econometric Analysis *mutually exclusive with ECON 140 and STAT 135	Introduction to problems of observation, estimation, and hypothesis testing in economics. This course covers the statistical theory for the linear regression model and its variants, with examples from empirical economics.	4	ECON 100A-100B or ECON 101A-101B or equivalent; Statistics 20, 21, 25, or 131A, or equivalent; and Mathematics 53 and 54, or equivalent; ECON 140, 141, and STAT 135 are mutually exclusive. Students may only take one of these three courses, and must select a second course from elsewhere on this list
EECS 127: Optimization Models in Engineering	This course offers an introduction to optimization models and their applications, ranging from machine learning and statistics to decision-making and control, with emphasis on numerically tractable problems, such as linear or constrained least-squares optimization.	4	EE 16A & 16B or consent of instructor

EL ENG 120: Signals and Systems	Continuous and discrete-time transform analysis techniques with illustrative applications. Linear and time-invariant systems, transfer functions. Fourier series, Fourier transform, Laplace and Z-transforms. Sampling and reconstruction. Solution of differential and difference equations using transforms. Frequency response, Bode plots, stability analysis. Illustrated by analysis of communication systems and feedback control systems.	4	EE 16A and 16B
EL ENG 123: Digital Signal Processing	Discrete time signals and systems: Fourier and Z transforms, DFT, 2-dimensional versions. Digital signal processing topics: flow graphs, realizations, FFT, chirp-Z algorithms, Hilbert transform relations, quantization effects, linear prediction. Digital filter design methods: windowing, frequency sampling, S-to-Z methods, frequency-transformation methods, optimization methods, 2-dimensional filter design.	4	EL ENG 120
EL ENG 129: Neural and Nonlinear Information Processing	Principles of massively parallel real-time computation, optimization, and information processing via nonlinear dynamics and analog VLSI neural networks, applications selected from image processing, pattern recognition, feature extraction, motion detection, data compression, secure communication, bionic eye, auto waves, and Turing patterns.	3	EL ENG 120 or consent of instructor
ESPM 174: Design and Analysis of Ecological Research	Surveys major designs and analyses for biological field and laboratory studies. Topics include data distributions; regression; analysis of variance; fixed and random effects; blocking, split plots, and repeated measures; maximum likelihood; Generalized Linear Models; basic computer programming. Relies on math to interpret and manipulate equations supported by computer simulations. Examples include population, ecosystem, behavioral, and evolutionary ecology.	4	One year calculus; one semester statistics or consent of instructor
IND ENG 115: Industrial and Commercial Data Systems	Design and implementation of databases, with an emphasis on industrial and commercial applications. Relational algebra, SQL, normalization. Students work in teams with local companies on a database design project. WWW design and queries.	3	Upper division standing
IND ENG 135: Applied Data Science with Venture Applications	This highly-applied course surveys a variety of key of concepts and tools that are useful for designing and building applications that process data signals of information. The course introduces modern open source, computer programming tools, libraries, and code samples that can be used to implement data applications. The mathematical concepts highlighted in this course include filtering, prediction, classification, decision-making, Markov chains, LTI systems, spectral analysis, and frameworks for learning from data. Each math concept is linked to implementation	3	Ability to write code in Python, and a probability or statistics course. This course is ideal for students who have taken CS/INFO/STAT C8

	using Python using libraries for math array functions (NumPy), manipulation of tables (Pandas), long term storage (SQL, JSON, CSV files), natural language (NLTK), and ML frameworks. **Student Learning Outcomes:** Students will be able to design and build data sample application systems that can interpret and use data for a wide range of real life applications across many disciplines and industries; implement these concepts within applications with modern open source CS tools; understand relevant mathematical concepts that are used in systems that process data.		
INFO 159: Natural Language Processing	This course introduces students to natural language processing and exposes them to the variety of methods available for reasoning about text in computational systems. NLP is deeply interdisciplinary, drawing on both linguistics and computer science, and helps drive much contemporary work in text analysis (as used in computational social science, the digital humanities, and computational journalism). We will focus on major algorithms used in NLP for various applications (part-of-speech tagging, parsing, coreference resolution, machine translation) and on the linguistic phenomena those algorithms attempt to model. Students will implement algorithms and create linguistically annotated data on which those algorithms depend.	4	CS 61B; CS 70, CS C100, MATH 55, STATS C100, STATS 134 or STATS 140; strong programming skills
INFO 190-1. Introduction to Data Visualization	This course introduces students to data visualization: the use of the visual channel for gaining insight with data, exploring data, and as a way to communicate insights, observations, and results with other people. The field of information visualization is flourishing today, with beautiful designs and applications ranging from journalism to marketing to data science. This course will introduce foundational principles and relevant perceptual properties to help students become discerning judges of data displayed visually. The course will also introduce key practical techniques and include extensive hands-on exercises to enable students to become skilled at telling stories with data using modern information visualization tools. Students will be asked to complete assignments before class, work together in small groups in class, and provide peer assessments. Grades will be based on assignments, quizzes, in class participation, peer assessment quality, 2 midterms, and a final project. The assignments for the course will together work towards building a coherent visualization that tells a story and is visible on the web.	3	This course is designed for upper division undergraduates who have an interest in design and in data. It is intended to accommodate students who have only a limited programming background, as well as those who are skilled with programming. For this reason, the only prerequisite is CS/Stat/Info 8 or equivalent. This course assumes students already have familiarity with basic data analysis and manipulation, and basic statistics. Students are encouraged but not required to have taken other courses from the introductory design sequence (one of DES INV 10- Discovering Design DES INV 15- Design Methodology, DES INV 21- Visual

			Communications & Sketching, CS 160 User Interface Design and Development), as well as other introductory data science and statistics courses. Graduate students will be accommodated only as space permits.
NUC ENG 175. Methods of Risk Analysis	Methodological approaches for the quantification of technological risk and risk based decision-making. Probabilistic safety assessment, human health risks, environmental and ecological risk analysis. Course Objectives: Introduce students to the meaning of risk, its mathematical formulations and how it is used to manage the societal impacts, including risk, cost and benefit, of modern technologies. Introduce students to the construction and quantification of fault and event trees. Introduce students to methods of consequence analysis, including multi-pathway exposure assessment. Introduce students to the principles of risk communication and risk perception. Course Outcomes: Construct and quantify fault and event trees for nuclear and other engineered systems including structures, systems and components. Quantify consequences of release scenarios, including multimedia exposure for toxic and radioactive materials in the biosphere. Apply risk based decision methods to nuclear and other engineered systems for the determination of cost/benefit and risk/benefit considerations. Communicate the results of risk analyses to multi-stakeholders Topics Covered: Definitions and measures of risk. Probabilistic safety assessment, fault and event trees, consequence analysis, risk integration. Environmental and public health risk assessment, source terms, environmental fate and transport, exposure assessment, health risk assessment. Ecological risk assessment. Risk based regulations, risk informed decision making. Risk management, decision trees, influence diagrams, multi-attribute decision theory. Uncertainty and variability in risk assessment and risk management. Case studies: nuclear systems, environmental systems and chemical systems	3	The course provides a general introduction to risk assessment, risk management and risk communication. Lower-division math, physics and chemistry, and an introduction to probability and statistics.

PHYSICS 151. Data Science and Bayesian Statistics for Physical Sciences	The goals of the course is to get acquainted with modern computational methods used in physical sciences, including numerical analysis methods, data science and Bayesian statistics. We will introduce a number of concepts that are useful in physical sciences at varying depth levels. We will cover main numerical methods used in physical sciences. Most of the statistical concepts will be Bayesian, emphasizing the concepts that have a connection to physical sciences, such as classical and statistical mechanics. We will focus on data science and data analysis applications that are often encountered in real world of physical sciences. We will review many of the machine learning concepts and methods.	4	Undergraduate students: PHY7 or PHY5 series, basic introduction to Python programming at the level of PHY77 or permission from instructor. Some knowledge of analytic mechanics and statistical physics at the level of PHY105 and PHY112 will be assumed. Graduate students: none.
STAT 135.* Concepts of Statistics	A comprehensive survey course in statistical theory and methodology. Topics include descriptive statistics, maximum likelihood estimation, non-parametric methods, introduction to optimality, goodness-of-fit tests, analysis of variance, bootstrap and computer-intensive methods and least squares estimation. The laboratory includes computer-based data-analytic applications to science and engineering.	4	ECON 140, 141, and STAT 135 are mutually exclusive. Students may only take one of these three courses, and must select a second course from elsewhere on this list; Course is not repeatable for credit.STAT 134 or STAT 140; and MATH 54, ELENG 16A, STAT 89A, MATH 110 or equivalent linear algebra. Strongly recommended corerequisite: STAT 133.
STAT 151A. Linear Modelling: Theory and Applications	Regression modeling • Visualization and diagnostics • Confidence Intervals and Hyp. Testing • Generalized Linear Models • Large numbers of predictors	4	STAT 135. Statistics 133 is recommended. STAT 135 implies other prerequisite courses (134 and its prerequisites). In particular, you must have had linear algebra, so you should be are familiar with: basic matrix operations, vector subspaces and projections, rank and invertibility of matrices, quadratic forms. This will form a core component of the course and is a real requirement.
STAT 152. Sampling Surveys	Estimation in probability surveys • Simple random samples • Stratified samples • Cluster samples • Complex samples • More advanced data analysis	4	STAT 134. Statistics 133 and 135 are recommended.
STAT 153. Introduction to Time Series	A time series is a set of numerical observations, each one being recorded at a specific time. Such data arise everywhere. This course aims to teach you how to analyze time series data (we will focus mostly on univariate time series data). There exist two main approaches to time series analysis: Time Domain approach and	4	This course is intended for students who have taken at least one elementary statistics course (some basic data analysis experience and familiarity with statistical

	Frequency Domain approach. Approximately, about 60% of the course will be on time domain methods and 40% on frequency domain methods. Tackling Trend and Seasonality, Stationarity and Stationary ARMA models, ARIMA and Seasonal ARIMA models, state space models, spectral distribution function and spectral density, Discrete Fourier Transform and the Periodogram, spectral density estimation.		notions such as correlation, maximum likelihood estimation, method of moments estimation, least squares, confidence intervals etc. is required) and one elementary probability course (at the level of STAT 134; familiarity is required with random variables, distribution functions, independence, uncorrelatedness, joint distributions etc). For the second part of the class (on frequency domain methods), some background on complex numbers is required.
STAT 158. The Design and Analysis of Experiments	An introduction to the design and analysis of experiments. This course covers planning, conducting, and analyzing statistically designed experiments with an emphasis on hands-on experience. Standard designs studied include factorial designs, block designs, latin square designs, and repeated measures designs. Other topics covered include the principles of design, randomization, ANOVA, response surface methodoloy, and computer experiments.	4	Statistics 134 and 135 or consent of instructor. Statistics 135 may be taken concurrently. Statistics 133 is recommended.
STAT 159. Reproducible and Collaborative Statistical Data Science	A project-based introduction to statistical data science. Through lectures, computational laboratories, readings, homeworks, and a group project, you will learn practical techniques and tools for producing statistically sound and appropriate, reproducible, and verifiable computational answers to scientific questions. The course emphasizes version control, testing, process automation, code review, and collaborative programming. Software tools include Bash, Git, Python, Jupyter and LATEX.	4	Statistics 133, Statistics 134, and Statistics 135 (or equivalent). Graduate standing is required to register for Statistics 259.

Probability

Course	Description	Units	Prerequisites
STAT 134. Concepts of Probability	An introduction to probability, emphasizing concepts and applications. Conditional expectation, independence, laws of large numbers. Discrete and continuous random variables. Central limit theorem. Selected topics such as the Poisson process, Markov chains, characteristic functions.	4	Mastery of the material in Appendices 1-4 of the text, fluency with calculus (derivatives and integrals) in two variables, and – these are crucial – clear logical reasoning and strong problem-solving skills. Test yourself on some practice

			problems.
STAT 140. Probability for Data Science	Prob 140 (formally Statistics 140 or STAT 140) is a probability course for undergraduates who have taken Data 8, have a math background, and wish to go deeper into the theory of data science. The emphasis on simulation and the bootstrap in Data 8 gives students a concrete sense of randomness and sampling variability. Prob 140 will capitalize on this. Because of the students' backgrounds, Prob 140 will move swiftly over basics, avoid approximations that are unnecessary when SciPy is at hand, and replace some of the routine calculus by symbolic math done in SymPy. This will create time to focus on the more demanding concepts that are part of the theoretical foundations of data science. Prob 140 contents have been selected based on consultation with faculty who regularly teach Stat and CS courses in advanced statistical topics including machine learning. Specifically, Prob 140 is intended to provide good preparation for CS 189 and Stat 154, with the goal that students should be able to move from Prob 140 to a machine learning class without necessarily having to take a semester of theoretical statistics in between. The class requires more mathematics than Stat 134, the Statistics department's standard course in classical probability theory; in addition to calculus, Prob 140 requires linear algebra. Computational power in Prob 140 allows students to solve problems that are intractable by other methods. Students also explore the standard mathematical theory graphically and by simulation, and thus develop a more firm grasp of the concepts than they might by using math alone.	4	Prob 140 is restricted to undergraduates who: ● Have not taken Stat 134; only one of Stat 134 and Prob 140 can be taken for credit ● Have had a year of calculus at the level of Math 1A-1B and preferably higher; Prob 140 involves some double integration and partial derivatives ● Have taken or are concurrently taking linear algebra in Math 54 or EE 16A or Math 110, or have taken an equivalent linear algebra course at another college ● Have a passing grade in Data 8 (P is fine if the course was taken P/NP); the EdX MOOC Data 8X does not satisfy this requirement
IND ENG 172. Probability and Risk Analysis for Engineers	This is an introductory probability course for students in engineering. It focuses mostly on random variables and their applications. Applications will be given in such areas as reliability theory, risk theory, inventory theory, financial models, computer science, and others. Note: This course is a statistics course and cannot be used to fulfill any engineering unit or elective requirements.	3	Mathematics 1A-1B or 16A-16B No credit will be given for this class if you have taken STAT 134.
EL ENG 126. Probability and Random Processes	This course covers the fundamentals of probability and random processes useful in fields such as networks, communication, signal processing, and control. Sample space, events, probability law. Conditional probability. Independence. Random variables. Distribution, density functions. Random vectors. Law of large numbers. Central limit theorem. Estimation and detection. Markov chains.	4	CS 70.

Modeling, Learning, and Decision-Making

Course	Description	Units	Prerequisites
COMPSCI 182. Designing, Visualizing, and Understanding Deep Networks	Deep Networks have revolutionized computer vision, language technology, robotics and control. They have growing impact in many other areas of science and engineering. They do not however, follow a closed or compact set of theoretical principles. In Yann Lecun's words they require "an interplay between intuitive insights, theoretical modeling, practical implementations, empirical studies, and scientific analyses." This course attempts to cover that ground.	4	Math 53 and Math 54 or equivalent; CS70 or Stat 134; CS61B or equivalent; CS189 (recommended)
COMPSCI 189. Introduction to Machine Learning	Theoretical foundations, algorithms, methodologies, and applications for machine learning. Topics may include supervised methods for regression and classication (linear models, trees, neural networks, ensemble methods, instance-based methods); generative and discriminative probabilistic models; Bayesian parametric learning; density estimation and clustering; Bayesian networks; time series models; dimensionality reduction; programming projects covering a variety of real-world applications.	4	**Prerequisites:** Mathematics 53 and 54; Computer Science 70 or consent of instructor **Credit Restrictions:** Students will receive no credit for Comp Sci 189 after taking Comp Sci 289A.
IND ENG 142. Introduction to Machine Learning & Data Analytics	This course introduces students to key techniques in machine learning and data analytics through a diverse set of examples using real datasets from domains such as e-commerce, healthcare, social media, sports, the Internet, and more. Through these examples, exercises in R, and a comprehensive team project, students will gain experience understanding and applying techniques such as linear regression, logistic regression, classification and regression trees, random forests, boosting, text mining, data cleaning and manipulation, data visualization, network analysis, time series modeling, clustering, principal component analysis, regularization, and large-scale learning.	3	IEOR 165 or equivalent course in statistics. Prior exposure to optimization is helpful but not strictly necessary. Some programming experience/literacy is expected
STAT 154. Modern Statistical Prediction & Machine Learning	Theory and practice of statistical prediction. Contemporary methods as extensions of classical methods. Topics: optimal prediction rules, the curse of dimensionality, empirical risk, linear regression and classification, basis expansions, regularization, splines, the bootstrap, model selection, classification and regression trees, boosting, support vector machines. Computational efficiency versus predictive performance. Emphasis on experience with real data and assessing statistical assumptions.	4	Mathematics 53 or equivalent; Mathematics 54, Electrical Engineering 16A, Statistics 89A, Mathematics 110 or equivalent linear algebra; Statistics 135 or equivalent; experience with some programming language. Recommended prerequisite: Mathematics 55 or equivalent exposure to counting arguments

Human Contexts and Ethics

Course	Description	Units	Prerequisites
AMERSTD/AFRICAM C134. Information Technology and Society	This course assesses the role of information technology in the digitalization of society by focusing on the deployment of e-government, e-commerce, e-learning, the digital city, telecommuting, virtual communities, internet time, the virtual office, and the geography of cyber space. The course will also discuss the role of information technology in the governance and economic development of society.	4	None?
INFO 188. Beyond the Data: Humans and Values	This course blends social and historical perspectives on data with ethics, law, policy, and case examples to help students understand current ethical and legal issues in data science and machine learning. Legal, ethical, and policy-related concepts addressed include: research ethics; privacy and surveillance; bias and discrimination; and oversight and accountability. These issues will be addressed throughout the lifecycle of data--from collection to storage to analysis and application. The course emphasizes strategies, processes, and tools for attending to ethical and legal issues in data science work. Course assignments emphasize researcher and practitioner reflexivity, allowing students to explore their own social and ethical commitments.	3	None?
ISF 100J. The Social Life of Computing	In this class, we will look at computing as a social phenomenon: to see it not just as a technology that transforms but to see it as a technology that has evolved, and is being put to use, in very particular ways, by particular groups of people. We will be doing this by employing a variety of methods, primarily historical and ethnographic, oriented around a study of practices. We will pay attention to technical details but ground these technical details in social organization (a term whose meaning should become clearer and clearer as the class progresses). We will study the social organization of computing around different kinds of hardware, software, ideologies, and ideas.	4	None?
STS C100 /ISF C100G / HIST C182. Introduction to Science, Technology, and Society: Human Contexts and Ethics of Data	This course provides an overview of the field of Science and Technology Studies (STS) as a way to study how our knowledge and technology shape and are shaped by social, political, historical, economic, and other factors. We will learn key concepts of the field (e.g., how technologies are understood and used differently in different communities) and apply them to a wide range of topics, including geography, history, environmental and information science, and others. Questions this course will address include: how are scientific facts constructed? How are values embedded in technical systems?	4	None?

BIO ENG 100. Ethics in Science and Engineering	The goal of this semester course is to present the issues of professional conduct in the practice of engineering, research, publication, public and private disclosures, and in managing professional and financial conflicts. The method is through historical didactic presentations, case studies, presentations of methods for problem solving in ethical matters, and classroom debates on contemporary ethical issues. The faculty will be drawn from national experts and faculty from religious studies, journalism, and law from the UC Berkeley campus.	3	For certain Domain Emphases, additional HCE options are available (see domain emphasis pages for more details): None?
CY PLAN 101. Introduction to Urban Data Analytics	This course (1) provides a basic intro to census and economic data collection, processing, and analysis; (2) surveys forecasting and modeling techniques in planning; (3) demonstrates the uses of real-time urban data and analytics; and (4) provides a socio-economic-political context for the smart cities movement, focusing on data ethics and governance.	4	For certain Domain Emphases, additional HCE options are available (see domain emphasis pages for more details): Upper division standing.
ESPM C167/PUB HLTH C160. Environmental Health and Development	The health effects of environmental alterations caused by development programs and other human activities in both developing and developed areas. Case studies will contextualize methodological information and incorporate a global perspective on environmentally mediated diseases in diverse populations. Topics include water management; population change; toxics; energy development; air pollution; climate change; chemical use, etc.	4	For certain Domain Emphases, additional HCE options are available (see domain emphasis pages for more details): None?

DOMAIN EMPHASES

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